INTERNATIONAL MONETARY FUND

REPUBLIC OF MOLDOVA

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AD HOC REVIEW UNDER THE EXTENDED CREDIT FACILITY; REQUEST FOR AUGMENTATION AND REPHASING OF ACCESS, MODIFICATION OF PERFORMANCE CRITERIA, AND COMPLETION OF THE INFLATION CONSULTATION UNDER THE EXTENDED CREDIT FACILITY AND EXTENDED FUND FACILITY ARRANGEMENTS—DEBT SUSTAINABILITY ANALYSIS

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Joint Bank-Fund Debt Sustainability Analysis	
Risk of external debt distress	Low
Overall risk of debt distress	Moderate
Granularity in the risk rating	Not applicable
Application of judgment	No

Moldova's risk of external debt distress remains low but overall risk of debt distress has been reclassified to moderate, down from low risk in the December 2021 Debt Sustainability Analysis (DSA).[1] This is because the present value (PV) of public debt-to-GDP ratio remains below the indicative threshold under the baseline scenario but exhibits a sustained breach of the threshold under a real GDP growth shock scenario. All the external debt indicators remain below the relevant indicative thresholds under the baseline and shock scenarios. Overall, public debt is considered sustainable with current debt carrying capacity assessed as strong,[2] despite the COVID pandemic-induced increase in public debt in 2020 coupled with envisaged increasing financing

[1] The previous DSA, dated December 8, 2021, accompanied Moldova's Staff Report for the 2021 Article IV Consultation and request for an arrangement under the ECF/EFF program, IMF Country Report No. 21/1.

[2] Moldova's Composite Indicator (CI) index, based on October 2021 WEO update and the World Bank's 2020 CPIA, indicates that the country's debt carrying capacity is strong (3.14).

needs to mitigate the economic and humanitarian fallout of the war in Ukraine, and the projected medium-term developmental spending needs. This assessment reflects a baseline which assumes that the authorities remain committed to prudent fiscal policy under the ECF/EFF program. In the long term, the financing mix is expected to shift from concessional to commercial borrowing by 2032. Private sector external debt remains relatively high, compared to peers, which could pose significant risks to external debt sustainability. The public debt trajectory remains vulnerable to risks, particularly from shocks to real GDP growth, calling for the need to broaden growth drivers. In addition, improving governance, enhancing the effectiveness of public spending, and strengthening the management of fiscal risks, including those stemming from state-owned enterprises, are priorities to contain public debt vulnerabilities.

PUBLIC DEBT COVERAGE

1. Moldova's public debt includes obligations of the public sector (central government, local authorities, and public entities). Public debt includes external and domestic obligations of the central government, including arrears to suppliers and guaranteed debt. Domestic debt includes debt of state and municipal enterprises, companies with full or majority public ownership, and of local public authorities with maturity of a year and above, as stipulated in Law No. 419 (2006) on Public Sector Debt, State Guarantees and State On-lending.[3] Debt contracted by the central government and central bank owed to the IMF is also covered.[4] The debt coverage is on the residency basis.

Text Table 1. Moldova: Public Debt Coverage

Public debt coverage

	Subsectors of the public sector	Check box
1	Central government	X
2	State and local government	X
3	Other elements in the general government	
4	o/w: Social security fund	X
5	o/w: Extra budgetary funds (EBFs)	
6	Guarantees (to other entities in the public and private sector, including to SOEs)	X
7	Central bank (borrowed on behalf of the government)	X
8	Non-guaranteed SOE debt	X

[3] PPG debt covers gross debt of the general government. SOE debt in Moldova is primarily driven by externally financed project loans that are on-lent by the central government to SOEs. Both on-lending to the private sector (operationalized through commercial banks) and to SOEs are part of public guarantees and are included in the central government debt. Non-guaranteed SOEs debt is also included in the public debt. Debt of SOEs with maturity longer than a year accounts for about 2 percent of GDP as of 2021. Staff and the authorities will define an agenda with concrete steps towards expanding debt coverage for local governments' debt, SOEs and PPPs to include all existing debt obligations. Due to the lack of data, information on PPPs is currently limited. The change in coverage complicates intertemporal analysis of PPG debt. The contingent liabilities shock from SOE debt is set at the default value of 2 percent of GDP to reflect risks associated with borrowing of SOEs majority owned by the state, while a contingent liability shock of 12 percent of GDP is meant to also capture risks from PPPs and SOEs that are partially owned by the state.

[4] Includes SDR allocation in 2021 (about US$236 million).

Text Table 1. Moldova: Public Debt Coverage (concluded)

Public debt coverage and the magnitude of the contingent liability tailored stress test

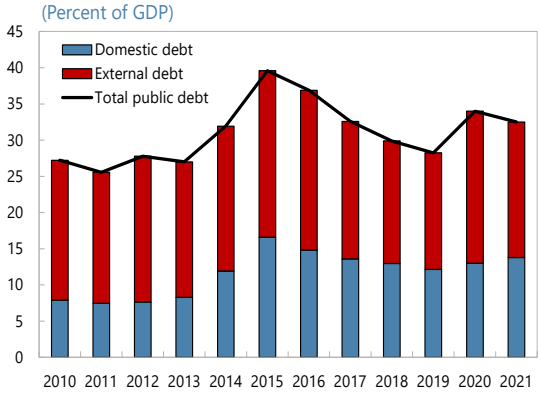
1	The country's coverage of public debt	The central, state, and local governments plus social security, central bank, government-guaranteed debt, non-guaranteed SOE		
		Default	Used for the analysis	Reasons for deviations from the default settings
2	Other elements of the general government not captured in 1.	0 percent of GDP	0.0	
3	SoE's debt (guaranteed and not guaranteed by the government) 1/	2 percent of GDP	2	
4	PPP	35 percent of PPP stock	0.0	
5	Financial market (the default value of 5 percent of GDP is the minimum value)	5 percent of GDP	10	
	Total (2+3+4+5) (in percent of GDP)		12.0	

1/ The default shock of 2% of GDP will be triggered for countries whose government-guaranteed debt is not fully captured under the country's public debt definition (1.). If it is already included in the government debt (1.) and risks associated with SoE's debt not guaranteed by the government is assessed to be negligible, a country team may reduce this to 0%.

BACKGROUND ON DEBT

2. Public debt declined in 2021 to 32.5 percent of GDP, from almost 34 percent of GDP in 2020, reflecting unwinding of the pandemic-induced increase in borrowing and the denominator effect from strong GDP. Public and publicly guaranteed (PPG) debt reached about 40 percent of GDP in 2015 up from about 27 percent of GDP in 2010 (Text Figure 1). A key driver of the spike in the public debt was the issuance of a state guarantee to the National Bank of Moldova (NBM) to provide emergency liquidity to the banking sector. During 2016–19, public debt was on a downward trend, returning towards the level seen in 2010. In 2019, after almost a year of a deadlock over concerns about governance and deteriorating democratic standards during the pre-election campaign, the EU and other donors resumed their financial assistance to Moldova. However, in 2020, COVID pandemic-induced borrowing pushed up public debt, driven by increases in both domestic and external debt (Text Table 2).[5] A rebound in strong economic activity and receding tailwinds from COVID-19 reduced government spending and borrowing in 2021, contributing to a decline in public debt to 32.5 percent of GDP. Currently, about two-third of PPG external debt is from multilateral institutions and about a third of PPG domestic marketable debt are long-term debt securities (government securities with maturity longer than one year). Other domestic marketable debt is mainly short-term and held by the banking system. In addition, the stock of domestic arrears to suppliers amounting to about MDL 30 million (0.01 percent of GDP) by end-2021 is included in domestic debt.

Text Figure 1. Evolution of Public Debt, 2010-2021
(Percent of GDP)



Sources: IMF WEO and IMF staff calculations.

[5] Although eligible, Moldova did not participate in the 2020–21 Debt Service Suspension Initiatives.

Text Table 2. Moldova: Decomposition of Public Debt and Debt Service by Creditor, 2021–2023 1/

(Millions of U.S. dollars, unless otherwise indicated)

	Debt Stock (end of period)			Debt Service					
	2021			2021	2022	2023	2021	2022	2023
		(Percent total debt)	(Percent GDP)				(Percent GDP)		
Total	4617.4	100.0	33.8	1938.4	1912.1	2007.6	14.2	13.7	13.5
External	2650.9	57.4	19.4	226.0	216.3	233.8	1.7	1.6	1.6
Multilateral creditors[2,3]	2532.1	54.8	18.5	154.4	182.1	224.6	1.1	1.3	1.5
IMF	832.0	18.0	6.1						
World Bank	813.5	17.6	5.9						
ADB/AfDB/IADB	0.0	0.0	0.0						
Other Multilaterals	886.6	19.2	6.5						
o/w: list largest two creditors	580.7	12.6	4.2						
EIB	422.2	9.1	3.1						
EBRD	158.5	3.4	1.2						
list of additional large creditors	0.0	0.0	0.0						
Bilateral Creditors[2]	118.8	2.6	0.9	60.5	34.2	9.2	0.4	0.2	0.1
Paris Club	111.8	2.4	0.8	23.7	28.3	9.2	0.2	0.2	0.1
o/w: list largest two creditors	68.8	1.5	0.5						
JICA	49.6	1.1	0.4						
Government of Russia	19.2	0.4	0.1						
list of additional large creditors	0.0	0.0	0.0						
Non-Paris Club	7.0	0.2	0.1	36.8	5.9	0.0	0.3	0.0	0.0
o/w: list largest two creditors	6.5	0.1	0.0						
Government of Romania	5.7	0.1	0.0						
Novo Gaming M Technologies GMBH	0.9	0.0	0.0						
list of additional large creditors	0.0	0.0	0.0						
Bonds	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Commercial creditors	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
o/w: list largest two creditors	0.0	0.0	0.0						
list of additional large creditors	0.0	0.0	0.0						
Other international creditors	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0
o/w: list largest two creditors	0.0	0.0	0.0						
list of additional large creditors	0.0	0.0	0.0						
Domestic	1966.5	42.6	14.4	1712.4	1695.7	1773.8	12.5	12.2	11.9
Held by residents, total	1966.4	42.6	14.4	1712.4	1695.7	1773.8	12.5	12.2	11.9
Held by non-residents, total	0.1	0.0	0.0						
T-Bills	1024.4	22.2	7.5	1424.5	1561.0	1544.0	10.4	11.2	10.4
Bonds	857.4	18.6	6.3	212.2	102.3	207.0	1.6	0.7	1.4
Loans*	84.6	1.8	0.6	75.6	32.4	22.8	0.6	0.2	0.2
Memo items:	103.7								
Collateralized debt[4]	n/a	n/a	n/a						
o/w: Related	n/a	n/a	n/a						
o/w: Unrelated	0.0	n/a	n/a						
Contingent liabilities	103.7		0.8	0.0	0.5	0.5	0.0	0.0	0.0
o/w: Public guarantees	103.7		0.8	0.0	0.5	0.5	0.0	0.0	0.0
o/w: Other explicit contingent liabilities[5]	n/a	n/a	n/a						
Nominal GDP	13672.4			13672.4	13954.1	14911.7			

1/ As reported by Country authorities according to their classification of creditors, including by official and commercial.

2/ Some public debt is not shown in the table due to [confidentiality clauses/capacity constraints].(Include for all creditor groups where applicable)

3/ Multilateral creditors" are simply institutions with more than one official shareholder and may not necessarily align with creditor classification under other IMF policies (e.g. Lending Into

4/ Debt is collateralized when the creditor has rights over an asset or revenue stream that would allow it, if the borrower defaults on its payment obligations, to rely on the asset or revenue stream to secure repayment of the debt. Collateralization entails a borrower granting liens over specific existing assets or future receivables to a lender as security against repayment of the loan. Collateral is "unrelated" when it has no relationship to a project financed by the loan. An example would be borrowing to finance the budget deficit, collateralized by oil revenue receipts. See the joint IMF-World Bank note for the G20 "Collateralized Transactions: Key Considerations for Public Lenders and Borrowers" for a discussion of issues raised by collateral.

5/ Includes other-one off guarantees not included in publicly guaranteed debt (e.g. credit lines) and other explicit contingent liabilities not elsewhere classified (e.g. potential legal claims, payments resulting from PPP arrangements).

* Debt service projection does not include data on interest payment on SOE and local authorities contracts

3. Similarly, Moldova's gross total external debt declined by about 10 percentage points of GDP in 2021 from about 74 percent of GDP in 2020, driven by reduced private external borrowing. In 2021, PPG external debt declined by 2.2 percentage points of GDP from 2020 of about 21 percent of

GDP, reflecting tailwinds of the base effect of higher external borrowing to mitigate the effects of the pandemic in 2020 compared to about 16 percent of GDP in 2019. The observed reduction of gross total external debt during 2016–19 largely reflects the decline in private external debt, which at above 40 percent of GDP since 2010[6], remains relatively elevated compared to peers. The strong appreciation of the nominal exchange rate by about 14 percent during this period amid renewed capital inflows also contributed to the reduction in gross external debt. The recent decline in private sector debt was due to a decrease in overseas borrowing by the banking sector, reflecting the tailwinds impact of the banking crisis in 2014.[7] Similar to other Central and Eastern European countries, while gross total external debt in Moldova is sizable at 63.6 percent of GDP in 2021, about half of the private debt are liabilities to direct investors (Moldovan foreign-owned companies borrowing from their parent companies abroad). Short-term debt of the non-bank sector is high as well, about one-third of non-bank debt, and consists of trade credits, arrears, and other debt liabilities, mostly for the import of natural resources. Foreign assets of the non-bank sector have shrunk in recent years but remain sizable (about 35 percent of gross nonbank external liabilities) and mainly held in the form of currency, deposits, and short-term loans (trade credits). Thus, while high private external debt in Moldova continues to pose risks to external debt sustainability, overall risk of total external debt distress is assessed as low because PPG external debt is held mainly by multilateral and bilateral donors and is mostly medium and long term and on concessional terms.

MACROECONOMIC OUTLOOK

4. The macroeconomic assumptions underlying the projections are consistent with the proposed augmentation of the ECF/EFF-supported program in the Policy Note. The main changes relative to the previous DSA of December 2021 include a downward revision to the 2020 GDP contraction and a significant recovery in 2021 (Text Table 3). Long run macroeconomic assumptions remain broadly unchanged. Economic performance is expected to remain solid over the medium term, with steady growth, moderate inflation, and a gradual narrowing of the current account deficit:

- **Real GDP.** 2021 real GDP growth was higher than envisaged, jumping to almost 14 percent, reflecting a stronger-than-expected recovery from the pandemic.[8] However, the nascent recovery in activity is envisaged to weaken in 2022, with real growth expected to decline to 0.3 percent, owing to the war in Ukraine and sanctions against Russia and Belarus. Spillovers from the war will impact the economy through a variety of channels in the near and medium-term: disruptions to trade depressing net

[6] Private external debt includes Moldovagaz debt owed to Gazprom for gas delivery to Moldova, contributing to a peak in private external debt at about 48 percent of GDP in 2024, with the total external debt reaching 75.3 percent of GDP.

[7] NBM is working continuously on improving the coverage of private sector debt. This explains the changes in historical debt numbers from period to period.

[8] In 2018, Moldova's National Bureau of Statistics (NBS) published revised GDP series for 2016 and 2017, based on new methodology to reflect: a) implementation of the UN's System of National Accounts 2008 (2008 SNA) and the European System of Accounts 2010 (ESA 2010); and b) statistical improvements regarding data sources and compilation methods. The changes were introduced with technical assistance from the Fund. As a result of the new methodology, the level of both nominal and real GDP was revised up by about 17 percent. The sizeable GDP revision implies a reduction in key macroeconomic ratios (including debt-to-GDP ratios).

exports, adverse confidence effects weighing on foreign and domestic investment and exerting pressure on FX market. Remittances could also be impacted, although the impact is unknown at this point. High energy costs, in particular the hike in gas tariffs, coupled with anticipated higher food prices and implied uncertainty are also expected to negatively affect aggregate demand. Over the medium-term, growth is expected to strengthen to an average of about 5 percent, driven by broad-based program-induced reforms including additional fiscal support envisioned to close Moldova's developmental needs.[9] The additional fiscal support envisioned under the program is expected to be directed to address critical human and infrastructure bottlenecks (which have been ringfenced by a "floor" in the development spending as part of the IMF-supported program).

- **Inflation.** Driven by rising food prices, adjustments in energy tariffs, and pent-up demand, inflation accelerated to 22.2percent (y/y) in March, well above both the NBM's target band and the consultation band under the program. The war in Ukraine is also putting pressure on the MDL which depreciated significantly since the intensification of the conflict, with pass-through expected to exert pressure on domestic prices (Box 1, main text). Recent developments in the gas market have further complicated the picture due to high tariff adjustments feeding into regulated prices, while the implied decrease in purchasing power and arising uncertainty might curtail aggregate demand and core inflation. For 2022, average inflation is projected to rise to 23.5 percent (y/y), before receding towards the inflation target at end-2023 and over the medium-term, where inflation is expected to remain broadly anchored around the National Bank's inflation target. The inflation outlook is subject to significant downside risks.

- **Fiscal.** The fiscal position improved in 2021, with the deficit falling to 2.6 percent of GDP from 5.3 percent in the previous year due to spending under-execution and buoyant revenues, including increased grant financing. However, the energy crisis and spillover effects of the war in Ukraine are expected to strain the government balance sheet in 2022, with the deficit expected to increase to 7.2 percent of GDP. Revenues are projected to fall as a share of GDP relative to the past two years (30.7 percent of GDP vis a vis 32 percent in 2021 and 31.4 percent in 2020) as the impact of trade frictions and reduced consumer and business confidence are felt. Meanwhile, additional spending allocations of 2.5 percent of GDP are required to cushion the impact of escalating energy prices on households and businesses, support households living standards in the face of high inflation and provide urgent humanitarian support to refugees fleeing Ukraine, raising fiscal deficit, and exacerbating external financing needs. These multiple crises are expected to delay the implementation of some reforms envisioned under the ECF/EFF arrangement. However, over the medium-term, improvements in revenue and spending efficiency and supporting governance reforms under the program are expected to support fiscal consolidation and improve debt sustainability. Measures to be taken to that effect include strengthening tax administration and improving tax compliance; streamlining of tax exemptions; improvements to the efficiency of capital spending by strengthening public investment management; strengthening the unified public wage system and improving control over the wage bill; and current expenditure rationalization. Given the shock to revenues, the crisis-driven expansion in

[9] Nevertheless, the output gap remains negative in the medium-term following the sharp decline in GDP of 8.3 percent in 2020.

necessary spending, and binding domestic financing constraints in 2022, total public debt is envisaged to peak at 43 percent of GDP in 2024 before declining to 40.6 percent in 2027.

- **External sector.** The current account deficit worsened to 11.7 percent of GDP in 2021, from about 8 percent of GDP in 2020, reflecting a strong rebound in economic activity and an uptick in the energy import bill. The current account deficit is expected to worsen further to 13 percent of GDP in 2022 due to reduced exports and a spike in commodity prices of imports, before narrowing to 8.5 percent of GDP by 2027; financed by strong capital and investment flows catalyzed by the ECF/EFF program. Despite the pandemic, the MDL strengthened in 2020–21, but is expected to weaken in 2022, reflecting an increase in imports coupled with subdued capital inflows due to the heightened uncertainty. At end-December 2021, gross international reserves stood at US$3.9 billion, up from US$3.78 billion recorded in 2020, to comprise about 173 percent of the Fund's composite reserve adequacy metric. Gross international reserves are expected to decline in 2022 due to a slump in capital inflows amid heightened uncertainty and an increase in commodity prices of imports but are expected to trend upward from 2023 and beyond, ensuring reserves adequacy.

- **External borrowing.** The DSA assumes that all contracted but undisbursed concessional loans will be fully disbursed over the medium-term as planned by the authorities. New borrowings, including commercial with Moldova having graduated from IDA-eligible borrowing in 2021, are projected to rise over the longer term to finance the country's high development needs. This plays a key role in the DSA and explains to some extent the upward shift in debt burden indicators after 2030, including for total external debt level indicators.[10] Moldova will continue to benefit from significant grant financing in the medium-term, driven in large part by the profile of the new concessional borrowing, leaving the grant element of new borrowing at about 40 percent. In the short-term, the external financing need—estimated at about US$578 million in 2022[11]—will be fully covered by the World Bank's Development Policy Operation (DPO) (US$150 million),[12] the European Commission (US$121 million), other development partners' disbursements (about US$100 million), and the augmented ECF/EFF disbursement of about US$208 million (SDR 149.45 million, of which SDR 99.90 million under the EFF and SDR 49.55 million under the ECF), all of which are to be disbursed for budget support.[13] In 2023–25, the remaining external financing gap of US$819 million will be financed mainly by the IMF disbursements (around US$550 million) under the ECF/EFF program, EU funds via macro-financial assistance to Moldova under a new financing package (about US$129 million), and the World Bank's DPO (US$140 million).

[10] While this assumption is not based on concrete borrowing plans in the longer-term, it reflects the baseline assumptions, under which Moldova will continue to borrow into the future to finance productive infrastructure investments.
[11] Exceptional financing.
[12] This includes exceptional financing from the IDA19 Crisis Response Window, as approved by the Board of Executive Directors in June 2021 and the anticipated approval of the Exceptional IDA19 Support to Moldova endorsed by IDA Deputies in April 2022.
[13] The authorities also received new SDR allocation of about US$236 million (SDR 165.3 million) which was channeled for budget support.

- **Public debt to GDP is expected to increase in 2022–27 with an increase in domestic and external borrowing.** An increase in public debt reflects the government increasing financing needs to mitigate the economic and humanitarian spillovers of the war in Ukraine, while also responding to rising energy cost and manage the still-emerging COVID variants' impact on livelihoods and the economy. In 2022, external debt is envisaged to rise by about 6 percentage points of GDP with financing from multilateral and bilateral sources. Domestic borrowing is assumed to be net zero in 2022, covering the costs of rolling over existing debt. Demand for government securities is expected to improve in 2023, allowing modest net financing of MDL 5 billion and thereafter we assume demand to resurge. The weighted average interest rate on short-term debt is assumed to be around 13 percent in 2022, falling to 6 percent in 2023 and declining thereafter over the medium-term to 4.5 percent.

Text Table 3. Moldova: Key Macroeconomic Assumptions
(DSA March 2022 vs DSA December 2021)

	2019-2021	2022-2027	2028-2032	2022-32
Real GDP growth (percent)				
Curent DSA	3.1	3.9	5.0	4.4
Previous DSA	1.4	5.2	5.0	5.1
Inflation (GDP deflator, in US dollar terms)				
Curent DSA	3.3	0.8	1.9	1.3
Previous DSA	3.1	0.8	1.9	1.3
Total Revenue (percent of GDP)[1]				
Curent DSA	30.4	31.6	33.0	32.2
Previous DSA	30.1	31.8	32.3	32.0
Current Account Deficit (non-interest, in percent of GDP)				
Curent DSA	8.9	9.6	6.6	8.2
Previous DSA	8.7	7.6	5.5	6.6

Source: Moldova authorities and Staff calculations.
[1] Total revenue, excluding grants.

5. The debt sustainability framework's realism tools suggest that the baseline projections are reasonable (Figures 3 and 4). There is a deviation in the evolution of the projection of both external and PPG debt to GDP ratios for the current and previous DSA vintages, and from the DSA from the 5 years past. In terms of projections, the current DSA deviates from the previous DSA mainly due to the incorporation of: (i) revised macroeconomic assumptions reflecting subdued growth in 2022 (a denominator) due to the war in Ukraine, and (ii) the looser fiscal policy going forward to support developmental needs and growth. For the **external public debt**, projected debt ratios over the medium term have risen owing to an expected increase in financing needs to provide social and economic support to cushion the effects of the war in Ukraine on Moldova, in line with 5-year ahead projected changes. In addition, higher financing cost (high interest rate) is also projected to increase debt ratios. The increase in the external debt ratio relative to the five-year historical change is due mostly to exchange rate movements, other residuals, and developments

in the current account and FDI.[14] For the **total public debt**, the increase in the projected debt ratios to GDP in the medium term is driven mainly by the widening primary balance and exchange rate movements which more than offsets the negative contribution to debt from strong real GDP growth, whereas the decline compared to the five-year historical debt ratios was due to real exchange rate factors and resilient real GDP growth. The difference over 2021–22 between the baseline growth projections and growth projections implied by the standard fiscal multipliers in Figure 4, reflects the scarring impact of the COVID pandemic in some sectors of the economy and impact of the war in Ukraine on confidence in the economy in 2022. Growth is expected to rebound in 2023 to 2 percent and to hover at an average of 5 percent in the medium-term. The projected 3-year adjustment in the primary deficit is similar to observed in historical data from peers with Fund-supported programs (Figure 4) and is consistent with the authorities' commitment with continued fiscal consolidation.

6. Public investment and growth. The ECF/EFF-supported program aims at augmenting public investment with growth-enhancing structural reforms, including SOEs reform, and an improvement of the business environment. Both are expected to crowd-in and enhance domestic and foreign private investments in support of the higher projected growth trajectory over the medium term (Figure 4).

COUNTRY CLASSIFICATION AND DETERMINATION OF SCENARIO STRESS TESTS

7. Moldova's debt carrying capacity is assessed to be strong, unchanged from the previous DSA. The composite indicator (CI), which captures the impact of several factors through a weighted average of an institutional indicator,[15] real GDP growth, remittances,[16] international reserves, and world growth, confirms that Moldova's debt carrying capacity is classified to be strong, which is unchanged from the previous two DSA rounds (Text Table 4).[17]

8. The DSA includes a stress test that follows standardized settings.[18] All the PPG external debt indicators remain below the policy relevant thresholds (Table 3 and Figure 1). The standardized stress test indicates a breach of public debt thresholds beyond 2028 (Table 4 and Figure 2). Moldova does not have prominent economic features such as significant reliance on commodity exports, market financing that would require additional tailored stress tests or other modules, nor severely affected by natural disasters. Regarding the contingent liability stress test, a shock of 12 percent of GDP is used. The

[14] The relatively large residuals in external financing mainly reflect a drawdown of reserves, caused by financing from the capital and financial accounts falling short of the current account deficit.
[15] The World Bank's Country Policy and Institutional Assessment (CPIA).
[16] Remittances for Moldova comprise of two Balance of Payments (BoP) accounts: compensation of employees and remittances.
[17] Moldova's Composite Indicator (CI) is 3.14, which corresponds to a strong debt-carrying capacity based on the IMF's October 2021 World Economic Outlook and World Bank's 2020 CPIA.
[18] The shock applied is Real GDP growth set to its historical average (10 years) minus one standard deviation, or the baseline projection minus one standard deviation, whichever is lower for the second and third years of the projection period.

severity of the shock was calibrated to the most recent domestic banking crisis event in 2016 that resulted in recapitalization of about 8.3 percent of GDP.

Text Table 4. Moldova: Debt Carrying Capacity and Applicable Thresholds

Debt Carrying Capacity and Thresholds

Country	Moldova
Country Code	**921**

Debt Carrying Capacity	**Strong**

Final	Classification based on current vintage	Classification based on the previous vintage	Classification based on the two previous vintages
Strong	Strong 3.14	Strong 3.11	Strong 3.23

Note: Until the April 2019 WEO vintage is released, the two previous vintages ago classification and corresponding score are based solely on the CPIA per the previous framework.

Applicable thresholds

APPLICABLE	
EXTERNAL debt burden thresholds	
PV of debt in % of	
Exports	240
GDP	55
Debt service in % of	
Exports	21
Revenue	23

APPLICABLE	
TOTAL public debt benchmark	
PV of total public debt in percent of GDP	70

EXTERNAL DEBT SUSTAINABILITY ANALYSIS

9. Under the baseline scenario and alternative scenarios, all external debt indicators continue to remain below their policy-relevant thresholds (Table 1, Figure 1). During 2022–27, new external financing will consist primarily of borrowing from multilateral and bilateral lenders, while commercial borrowing is expected to start playing a larger role in the long-term from 2032, reaching about 65 percent of total public sector borrowing by 2042. The present value of PPG external debt is projected at 16 percent of GDP in 2022, decreasing to 8.9 percent by 2032. The ratio will remain well below the 55 percent threshold under the baseline scenario throughout the projection period. Similarly, debt service indicators remain below their respective thresholds. Improvements in debt-management practices envisaged under the authorities' reforms supported by the ECF/EFF program will give further resilience to shocks affecting debt service needs. A tailored stress test for the contingent liability shock also does not cause any breach of

relevant thresholds. Under the most extreme scenario, most PPG debt indicators show increase in their values but remain well below their indicative thresholds. These outcomes hinge on continued prudent fiscal policy and sound macroeconomic and debt management policies envisaged under the ECF/EFF program.

10. While the external risk rating is determined by the PPG external debt, large private external debt poses some potential rollover risks. In the baseline scenario, private external debt to GDP ratio is expected to remain broadly stable at about 45 percent of GDP in 2022, but it is envisaged to decline to 43 percent of GDP by 2027, and trend lower to about 35 percent of GDP in 2032 supported by the strong real growth and by a larger contribution of FDIs and other non-debt creating flows to the current account financing. Potential rollover risks from a significant stock of private sector debt are partly mitigated by private sector savings abroad.

PUBLIC DEBT SUSTAINABILITY ANALYSIS

11. Under the baseline, the indicators of the overall public debt burden are below the benchmark threshold, but risks remain. Under the baseline, the PV of total PPG debt-to-GDP ratio is projected to increase initially to 31.6 percent of GDP in 2022, from 26.9 percent of GDP in 2020, and is expected to stabilize below 35 percent of GDP over the medium-term, remaining below the 70 percent benchmark in the medium-to-long term (Figure 2).

12. The ratio of PV of total PPG debt-to-GDP ratio exceeds its threshold under the real GDP growth shock. Over the long-term, under the real GDP growth shock scenario it remains elevated, breaching the threshold starting from 2028, as the country accumulates higher debt to finance larger fiscal and current account deficits. Such a scenario highlights the risks to debt sustainability faced by the authorities in the absence of needed reforms. A significant contingent liabilities shock (to SOEs and PPPs and/or financial market distress) would also increase debt levels notably, though such risks are difficult to quantify accurately due to lack of quality data on SOEs and PPPs. The authorities' planned structural reforms underpinned by the ECF/EFF program are expected to improve revenues, increase spending efficiency, and boost growth, thereby mitigating further risk of unsustainable debt. As such, Moldova's overall debt trajectory is projected to remain sustainable despite the expansion in fiscal spending under the ECF/EFF program that is needed to achieve pressing developmental goals as well as to support the post-Covid recovery and provide economic and humanitarian assistance due to the war in Ukraine.

RISK RATING AND VULNERABILITIES

13. Moldova's risk of external debt distress remains low but overall risk of debt distress has been reclassified to moderate, down from low risk in the previous DSA assessments.

- **Debt burden indicators for PPG external debt remain well below the indicative debt thresholds, under both the standardized and alternative stress tests.** However, significant private external debt poses potential roll-over risks.

- **Moldova's overall public debt dynamics is also projected to remain on a sustainable path under the baseline scenario but remain subject to heightened vulnerabilities under the stress scenario.** Stress tests indicate that Moldova's debt sustainability is sensitive to the growth shock scenario.

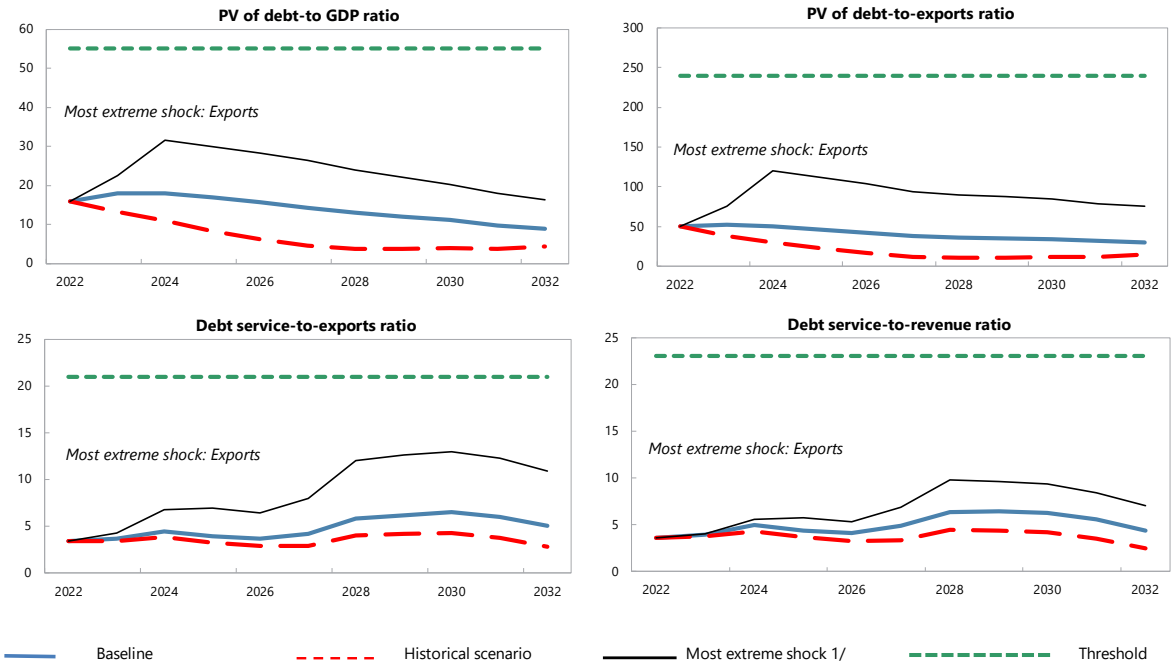
- **Risks to the debt outlook are tilted to the downside.** Given sensitivity of the debt projections to growth, adverse spillovers from a protracted war in Ukraine coupled with a potential resurgence of external and domestic containment measures linked to another wave of COVID-19 variant presents significant risk to the outlook on debt.

- **Mitigating debt risk requires sound macroeconomic management and sustained structural reforms**. The authorities should continue to maintain fiscal discipline and fiscal risk management. As the country is expected to accumulate significant debt to finance larger fiscal and current account deficits, strong and sustained political commitment is needed to deliver on fiscal policy with prudent deficit levels, revenue mobilization, and improved debt and fiscal risk management are imperative to safeguard Moldova's debt sustainability. In the near and medium-term, authorities should rely on concessional financing consistent with the ECF/EFF-supported program and further strengthen public investment management to finance priority projects with high growth and social returns. Contingent liability risks—including from SOEs—should be monitored carefully. Advancing structural reforms remain key to increasing the economy's growth potential, reducing vulnerability to shocks. Furthermore, development of the domestic debt market could further strengthen the outlook for debt sustainability, especially considering the expected transition of financing mix from concessional to commercial over the long term. As part of the ECF/EFF-supported program, steps will also be undertaken to strengthen debt and cash management to enable a broader range of longer maturity debt instruments to lengthen the maturity of domestic debt and deepen the secondary government securities market. Moldova will also seek to further strengthen debt management by developing domestic debt market and broaden secondary dealer system. These will help to reduce the PPG domestic debt roll-over and interest rate risks.

AUTHORITIES' VIEWS

14. The authorities broadly agreed with the staff's assessment of Moldova's public debt situation and recommendations on debt management policy. They broadly concurred with the staff's assessment of debt composition, projections, risk ratings and distress level. They recognize that maintaining fiscal discipline and managing fiscal risks is critical to ensure fiscal and debt sustainability. While they have been making progress on debt management and development of domestic debt market, they highlighted the need for further improvement in debt statistics and debt management frameworks by making full use of IMF technical assistance and training resources.

Figure 1. Moldova: Indicators of Public Guaranteed External Debt Under Alternative Scenarios, 2022–2032



Customization of Default Settings	Size	Interactions
Tailored Stress		
Combined CL	Yes	
Natural disaster	n.a.	n.a.
Commodity price	n.a.	n.a.
Market financing	Yes	No

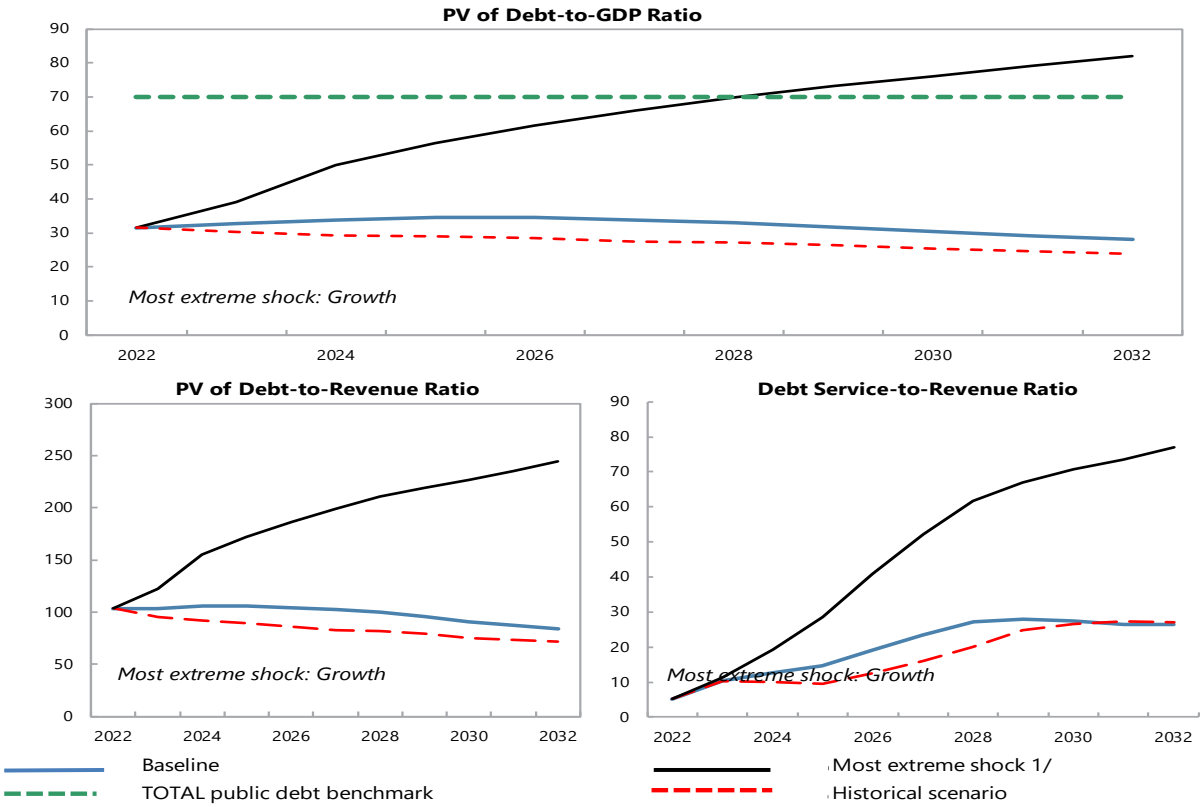
Note: "Yes" indicates any change to the size or interactions of the default settings for the stress tests. "n.a." indicates that the stress test does not apply.

Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG MLT debt	100%	
Terms of marginal debt		
Avg. nominal interest rate on new borrowing in USD	2.5%	2.5%
USD Discount rate	5.0%	5.0%
Avg. maturity (incl. grace period)	20	20
Avg. grace period	3	3

* Note: All the additional financing needs generated by the shocks under the stress tests are assumed to be covered by PPG external MLT debt in the external DSA. Default terms of marginal debt are based on baseline 10-year projections.

Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in or before 2032. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

Figure 2. Moldova: Indicators of Public Debt Under Alternative Scenarios, 2022–2032



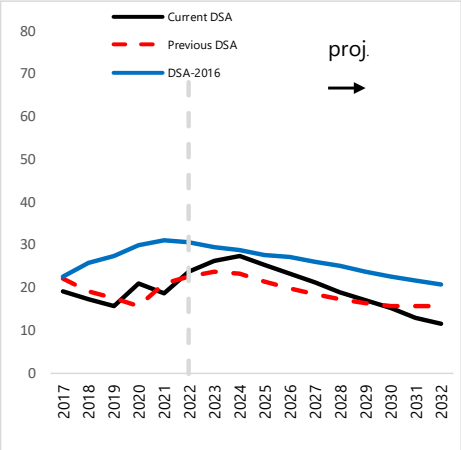
Borrowing assumptions on additional financing needs resulting from the stress tests*	Default	User defined
Shares of marginal debt		
External PPG medium and long-term	23%	23%
Domestic medium and long-term	39%	39%
Domestic short-term	38%	38%
Terms of marginal debt		
External MLT debt		
Avg. nominal interest rate on new borrowing in USD	2.5%	2.5%
Avg. maturity (incl. grace period)	20	20
Avg. grace period	3	3
Domestic MLT debt		
Avg. real interest rate on new borrowing	0.2%	0.2%
Avg. maturity (incl. grace period)	3	3
Avg. grace period	1	1
Domestic short-term debt		
Avg. real interest rate	-0.3%	-0.3%

* Note: The public DSA allows for domestic financing to cover the additional financing needs generated by the shocks under the stress tests in the public DSA. Default terms of marginal debt are based on baseline 10-year projections.
Sources: Country authorities; and staff estimates and projections.
1/ The most extreme stress test is the test that yields the highest ratio in or before 2032. The stress test with a one-off breach is also presented (if any), while the one-off breach is deemed away for mechanical signals. When a stress test with a one-off breach happens to be the most exterme shock even after disregarding the one-off breach, only that stress test (with a one-off breach) would be presented.

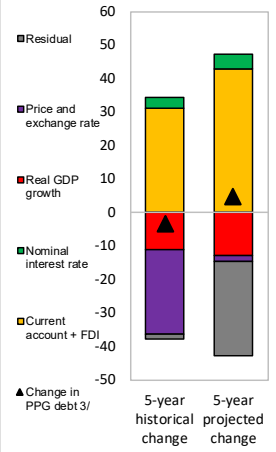
Figure 3. Moldova: Drivers of Debt Dynamics – Baseline Scenario

External debt

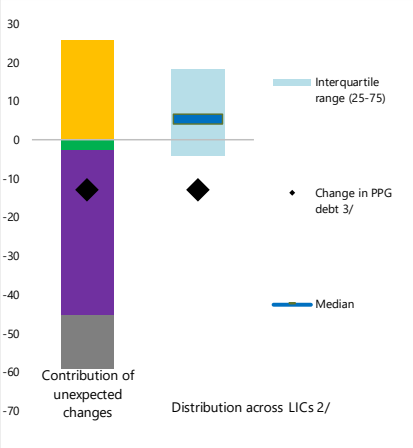
Gross Nominal PPG External Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)

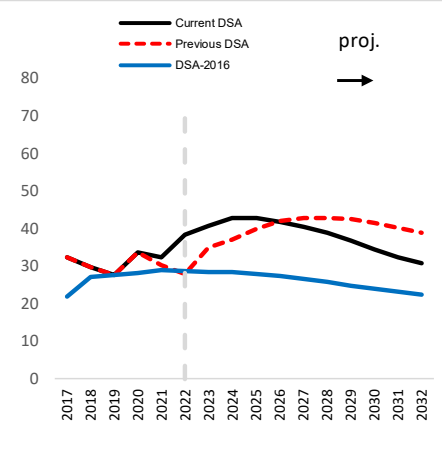


Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)

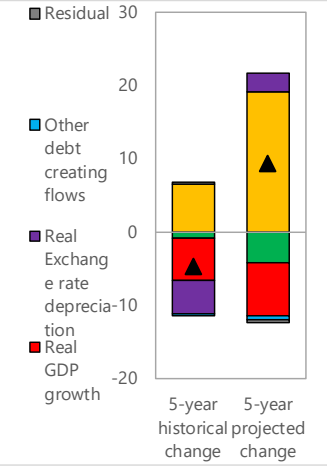


Public debt

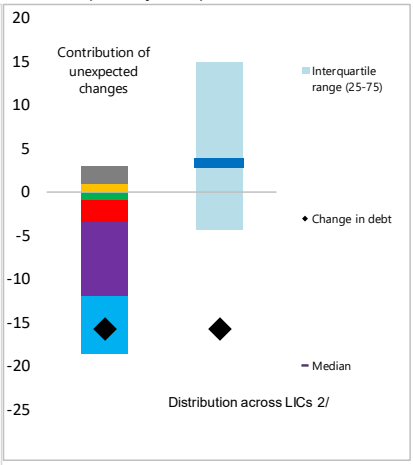
Gross Nominal Public Debt
(in percent of GDP; DSA vintages)



Debt-creating flows
(percent of GDP)



Unexpected Changes in Debt 1/
(past 5 years, percent of GDP)



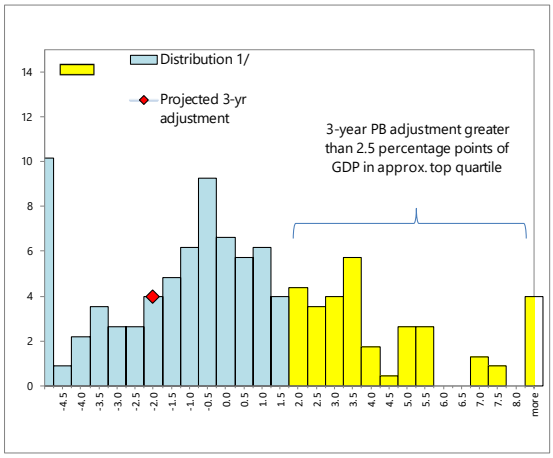
1/ Difference between anticipated and actual contributions on debt ratios.

2/ Distribution across LICs for which LIC DSAs were produced.

3/ Given the relatively low private external debt for average low-income countries, a ppt change in PPG external debt should be largely explained by the drivers of the external debt dynamics equation.

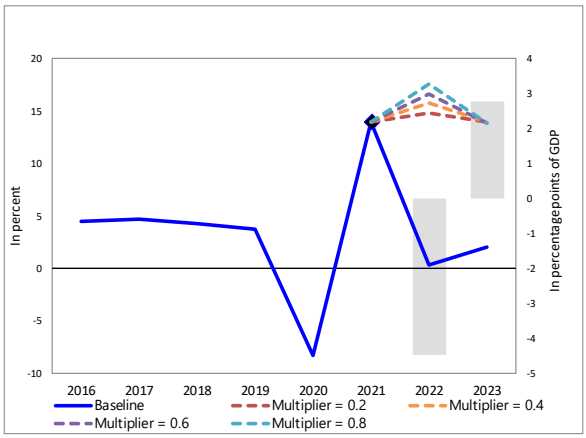
Figure 4. Moldova: Realism Tools

3-Year Adjustment in Primary Balance
(Percentage points of GDP)



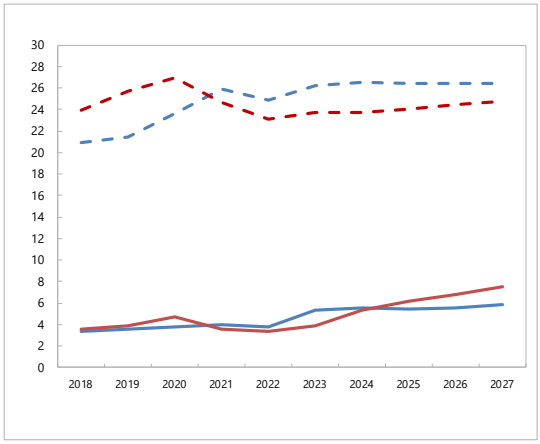
1/ Data cover Fund-supported programs for LICs (excluding emergency financing) approved since 1990. The size of 3-year adjustment from program inception is found on the horizontal axis; the percent of sample is found on the vertical axis.

Fiscal Adjustment and Possible Growth Paths 1/

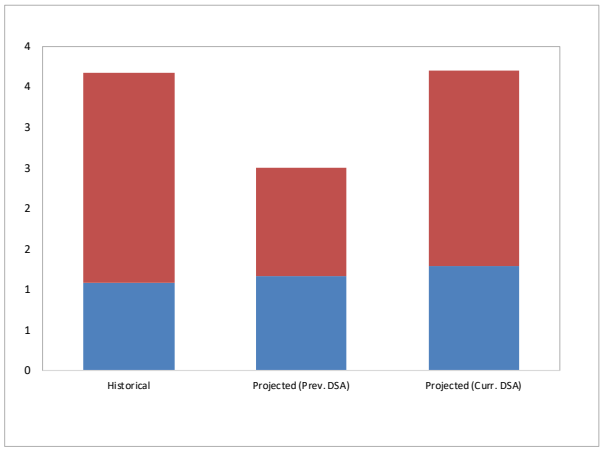


1/ Bars refer to annual projected fiscal adjustment (right-hand side scale) and lines show possible real GDP growth paths under different fiscal multipliers (left-hand side scale).

Public and Private Investment Rates
(percent of GDP)



Contribution to Real GDP growth
(percent, 5-year average)



Gov. Invest. - Prev. DSA
Priv. Invest. - Prev. DSA
Gov. Invest. - Curr. DSA
Priv. Invest. - Curr. DSA
Contribution of other factors
Contribution of government capital

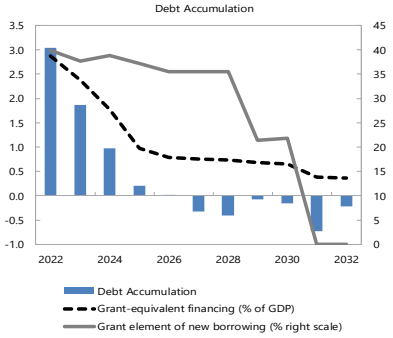
Table 1. Moldova: External Debt Sustainability Framework, Baseline Scenario, 2021–2042

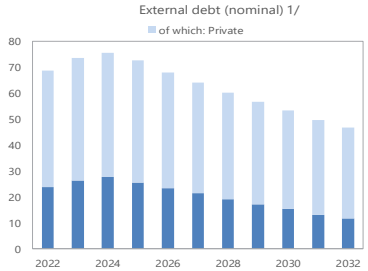
(in percent of GDP, unless otherwise indicated)

(In percent of GDP, unless otherwise indicated)

	Actual	Projections								Average 8/	
	2021	2022	2023	2024	2025	2026	2027	2032	2042	Historical	Projections
External debt (nominal) 1/	**63.6**	**68.7**	**73.6**	**75.3**	**72.6**	**67.9**	**64.0**	**46.7**	**36.4**	**70.8**	**62.6**
of which: public and publicly guaranteed (PPG)	**18.7**	**23.7**	**26.2**	**27.5**	**25.4**	**23.3**	**21.4**	**11.6**	**10.4**	**19.7**	**20.3**
Change in external debt	-9.9	5.1	4.9	1.8	-2.8	-4.6	-3.9	-2.8	-0.7		
Identified net debt-creating flows	**-1.6**	**12.3**	**9.2**	**5.0**	**4.3**	**3.9**	**3.6**	**1.9**	**-0.2**	**2.2**	**4.5**
Non-interest current account deficit	**11.1**	**12.4**	**11.1**	**9.5**	**8.8**	**8.2**	**7.4**	**6.0**	**3.2**	**6.9**	**8.2**
Deficit in balance of goods and services	27.3	29.9	30.0	28.9	27.4	26.1	25.1	19.3	11.3	**26.9**	**25.0**
Exports	30.7	31.7	34.2	35.7	36.2	36.8	37.9	29.1	17.1		
Imports	58.0	61.6	64.2	64.7	63.6	62.9	63.1	48.3	28.4		
Net current transfers (negative = inflow)	-12.9	-12.0	-12.4	-12.5	-11.9	-11.3	-11.0	-7.9	-4.0	**-13.3**	**-10.6**
of which: official	-0.4	-0.9	-0.6	-0.6	-0.5	-0.5	-0.5	-0.4	-0.2		
Other current account flows (negative = net inflow)	-3.3	-5.6	-6.5	-7.0	-6.7	-6.6	-6.7	-5.4	-4.1	**-6.7**	**-6.2**
Net FDI (negative = inflow)	**-1.7**	**-0.5**	**-1.5**	**-1.5**	**-1.6**	**-1.8**	**-1.8**	**-2.3**	**-2.3**	**-2.2**	**-1.8**
Endogenous debt dynamics 2/	**-10.9**	**0.4**	**-0.5**	**-2.9**	**-2.9**	**-2.5**	**-2.1**	**-1.8**	**-1.1**	**-2.4**	**-1.8**
Contribution from nominal interest rate	0.6	0.6	0.9	1.1	0.8	0.9	1.1	0.5	0.6		
Contribution from real GDP growth	-8.6	-0.2	-1.4	-4.1	-3.8	-3.4	-3.2	-2.3	-1.7		
Contribution from price and exchange rate changes	-2.9	...	...	...	...	...	...	...	...		
Residual 3/	**-8.3**	**-7.2**	**-4.3**	**-3.2**	**-7.1**	**-8.5**	**-7.4**	**-4.7**	**-0.5**	**-2.2**	**-6.1**
of which: exceptional financing	-1.7	-2.2	-1.1	-0.6	0.0	0.0	0.0	0.0	0.0		
Sustainability indicators											
PV of PPG external debt-to-GDP ratio	**13.1**	**16.0**	**17.9**	**18.0**	**16.9**	**15.7**	**14.3**	**8.9**	**10.0**		
PV of PPG external debt-to-exports ratio	**42.7**	**50.4**	**52.4**	**50.4**	**46.6**	**42.6**	**37.7**	**30.5**	**58.3**		
PPG debt service-to-exports ratio	**5.4**	**3.4**	**3.6**	**4.4**	**3.9**	**3.6**	**4.2**	**5.0**	**13.2**		
PPG debt service-to-revenue ratio	**5.3**	**3.6**	**4.0**	**5.0**	**4.4**	**4.1**	**4.9**	**4.4**	**6.8**		
Gross external financing need (Million of U.S. dollars)	4674.6	4892.9	5013.8	5199.2	5547.7	5995.4	6165.1	7424.5	11902.5		
Key macroeconomic assumptions											
Real GDP growth (in percent)	13.9	0.3	2.0	5.8	5.4	5.0	5.0	5.0	5.0	**3.6**	**4.4**
GDP deflator in US dollar terms (change in percent)	4.1	0.7	-2.4	-0.9	2.6	2.5	2.3	1.9	1.9	**1.9**	**1.3**
Effective interest rate (percent) 4/	0.9	1.0	1.3	1.6	1.2	1.3	1.7	1.1	1.8	**1.0**	**1.3**
Growth of exports of G&S (US dollar terms, in percent)	30.3	4.3	7.5	9.5	9.5	9.5	10.6	1.5	1.5	**5.2**	**5.3**
Growth of imports of G&S (US dollar terms, in percent)	33.6	7.3	3.7	5.6	6.4	6.5	7.7	1.5	1.5	**4.8**	**4.1**
Grant element of new public sector borrowing (in percent)	...	39.8	37.7	38.8	37.2	35.5	35.5	0.0	0.0	**...**	**27.6**
Government revenues (excluding grants, in percent of GDP)	31.0	29.6	31.0	31.6	32.2	32.6	32.5	33.1	33.1	**29.5**	**32.2**
Aid flows (in Million of US dollars) 5/	272.2	397.7	329.4	235.5	220.2	225.9	234.8	91.2	107.8		
Grant-equivalent financing (in percent of GDP) 6/	...	2.9	2.4	1.8	1.0	0.8	0.8	0.4	0.2	**...**	**1.1**
Grant-equivalent financing (in percent of external financing) 6/	...	49.4	47.7	45.7	53.9	58.5	58.1	31.3	8.2	**...**	**48.4**
Nominal GDP (Million of US dollars)	13,672	13,811	13,752	14,412	15,590	16,779	18,020	25,319	49,988		
Nominal dollar GDP growth	18.6	1.0	-0.4	4.8	8.2	7.6	7.4	7.0	7.0	**5.6**	**5.8**
Memorandum items:											
PV of external debt 7/	57.9	60.9	65.3	65.9	64.0	60.3	57.0	44.0	35.9		
In percent of exports	188.7	192.3	190.8	184.3	176.8	163.7	150.2	151.4	210.3		
Total external debt service-to-exports ratio	69.5	71.3	69.9	70.2	69.5	70.0	66.0	75.7	116.2		
PV of PPG external debt (in Million of US dollars)	1790.5	2206.4	2464.0	2598.4	2628.6	2631.2	2576.7	2244.8	4982.5		
(PVt-PVt-1)/GDPt-1 (in percent)		3.0	1.9	1.0	0.2	0.0	-0.3	-0.2	0.7		
Non-interest current account deficit that stabilizes debt ratio	21.0	7.2	6.2	7.7	11.6	12.8	11.3	8.8	3.9		

Definition of external/domestic debt	Residency-based
Is there a material difference between the two criteria?	Yes





Sources: Country authorities; and staff estimates and projections.

1/ Includes both public and private sector external debt.

2/ Derived as [r - g - ρ(1+g) + εα (1+r)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, ρ = growth rate of GDP deflator in U.S. dollar terms, ε=nominal appreciation of the local currency, and α= share of local currency-denominated external debt in total external debt.

3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.

4/ Current-year interest payments divided by previous period debt stock.

5/ Defined as grants, concessional loans, and debt relief.

6/ Grant-equivalent financing includes grants provided directly to the government and through new borrowing (difference between the face value and the PV of new debt).

7/ Assumes that PV of private sector debt is equivalent to its face value.

8/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

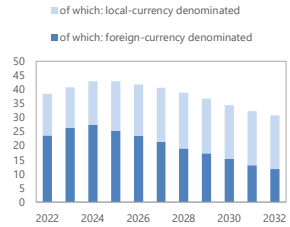
Table 2. Moldova: Public Sector Debt Sustainability Framework, Baseline Scenario, 2021–2042

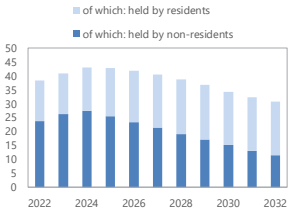
(in percent of GDP, unless otherwise indicated)

	Actual	Projections								Average 6/	
	2021	2022	2023	2024	2025	2026	2027	2032	2042	Historical	Projections
Public sector debt 1/	**32.5**	**38.4**	**40.8**	**43.0**	**42.8**	**41.8**	**40.6**	**30.7**	**23.0**	**30.7**	**38.2**
of which: external debt	**18.7**	**23.7**	**26.2**	**27.5**	**25.4**	**23.3**	**21.4**	**11.6**	**10.4**	**19.7**	**20.3**
Change in public sector debt	-1.3	5.9	2.4	2.1	-0.1	-1.0	-1.3	-1.6	-0.6		
Identified debt-creating flows	**-3.1**	**5.7**	**2.4**	**2.1**	**-0.1**	**-0.3**	**-0.2**	**-0.9**	**-0.6**	**1.0**	**0.5**
Primary deficit	**1.8**	**6.3**	**3.5**	**3.8**	**3.0**	**2.5**	**2.1**	**0.9**	**0.5**	**1.2**	**2.5**
Revenue and grants	32.0	30.6	31.8	32.1	32.7	33.0	33.0	33.5	33.3	**30.6**	**32.7**
of which: grants	1.0	0.9	0.8	0.4	0.5	0.5	0.5	0.4	0.2		
Primary (noninterest) expenditure	33.8	36.9	35.3	35.9	35.7	35.5	35.1	34.4	33.8	**31.8**	**35.2**
Automatic debt dynamics	**-4.9**	**-0.4**	**-1.0**	**-1.7**	**-3.1**	**-2.7**	**-2.3**	**-1.8**	**-1.1**		
Contribution from interest rate/growth differential	-5.0	-2.1	-1.3	-2.5	-2.9	-2.6	-2.3	-1.6	-0.9		
of which: contribution from average real interest rate	-0.9	-2.0	-0.6	-0.2	-0.7	-0.6	-0.3	-0.1	0.2		
of which: contribution from real GDP growth	-4.1	-0.1	-0.8	-2.2	-2.2	-2.0	-2.0	-1.5	-1.1		
Contribution from real exchange rate depreciation	0.2	1.7	0.3	0.8	-0.1	-0.1	-0.1	-0.2	-0.2		
Other identified debt-creating flows	**-0.1**	**-0.2**	**-0.1**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.0**	**0.8**	**-0.1**
Privatization receipts (negative)	-0.1	-0.2	-0.1	0.0	0.0	0.0	0.0	0.0	0.0		
Recognition of contingent liabilities (e.g., bank recapitalization)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Debt relief (HIPC and other)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Other debt creating or reducing flow (please specify)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		
Residual	**1.9**	**0.2**	**0.0**	**0.0**	**0.0**	**-0.7**	**-1.0**	**-0.7**	**0.0**	**-0.2**	**-0.7**
Sustainability indicators											
PV of public debt-to-GDP ratio 2/	**26.9**	**31.6**	**32.9**	**33.8**	**34.6**	**34.5**	**33.8**	**28.2**	**22.8**		
PV of public debt-to-revenue and grants ratio	**84.1**	**103.4**	**103.4**	**105.5**	**105.9**	**104.4**	**102.3**	**84.2**	**68.5**		
Debt service-to-revenue and grants ratio 3/	**8.1**	**5.2**	**10.3**	**12.8**	**14.7**	**19.3**	**23.4**	**26.5**	**20.2**		
Gross financing need 4/	4.4	7.7	6.7	7.9	7.8	8.8	9.8	9.7	7.2		
Key macroeconomic and fiscal assumptions											
Real GDP growth (in percent)	13.9	0.3	2.0	5.8	5.4	5.0	5.0	5.0	5.0	**3.6**	**4.4**
Average nominal interest rate on external debt (in percent)	0.9	0.7	1.1	2.3	0.3	0.8	1.8	0.7	4.6	**1.1**	**1.2**
Average real interest rate on domestic debt (in percent)	-3.2	-9.5	-1.9	-1.9	-1.9	-1.6	-1.4	-0.7	-1.7	**0.2**	**-2.0**
Real exchange rate depreciation (in percent, + indicates depreciation)	0.9	...	...	...	...	...	...	...	...	**0.8**	**...**
Inflation rate (GDP deflator, in percent)	6.3	12.7	6.0	5.5	5.5	5.5	5.5	5.0	5.0	**5.8**	**6.0**
Growth of real primary spending (deflated by GDP deflator, in percent)	7.5	9.4	-2.2	7.5	4.7	4.5	3.9	4.5	4.8	**4.1**	**4.6**
Primary deficit that stabilizes the debt-to-GDP ratio 5/	3.1	0.4	1.1	1.7	3.1	3.5	3.4	2.5	1.1	**1.4**	**2.7**
PV of contingent liabilities (not included in public sector debt)	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0		

Definition of external/domestic debt	**Residency-based**
Is there a material difference between the two criteria?	**Yes**

Public sector debt 1/





Sources: Country authorities; and staff estimates and projections.

1/ Coverage of debt: The central, state, and local governments plus social security, central bank, government-guaranteed debt, spending arrears, non-guaranteed majority owned SOE debt. Definition of external debt is Residency-based.

2/ The underlying PV of external debt-to-GDP ratio under the public DSA differs from the external DSA with the size of differences depending on exchange rates projections.

3/ Debt service is defined as the sum of interest and amortization of medium and long-term, and short-term debt.

4/ Gross financing need is defined as the primary deficit plus debt service plus the stock of short-term debt at the end of the last period and other debt creating/reducing flows.

5/ Defined as a primary deficit minus a change in the public debt-to-GDP ratio ((-): a primary surplus), which would stabilizes the debt ratio only in the year in question.

6/ Historical averages are generally derived over the past 10 years, subject to data availability, whereas projections averages are over the first year of projection and the next 10 years.

Table 3. Moldova: Sensitivity Analysis for Key Indicators of Public and Publicly Guaranteed External Debt, 2022–2032

(in percent)

	Projections 1/										
	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032
PV of debt-to GDP ratio											
Baseline	16	18	18	17	16	14	13	12	11	10	9
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2032 2/	16	13	11	8	6	5	4	4	4	4	4
B. Bound Tests											
B1. Real GDP growth	16	20	22	20	19	17	16	15	13	12	11
B2. Primary balance	16	18	19	18	17	15	14	13	12	11	10
B3. Exports	16	22	32	30	28	26	24	22	20	18	16
B4. Other flows 3/	16	20	23	22	20	19	17	16	14	13	11
B5. Depreciation	16	23	15	14	12	11	10	9	9	7	7
B6. Combination of B1-B5	16	24	24	22	21	19	17	16	15	13	12
C. Tailored Tests											
C1. Combined contingent liabilities	16	20	21	20	20	19	18	17	16	15	14
Threshold	55	55	55	55	55	55	55	55	55	55	55
PV of debt-to-exports ratio											
Baseline	50	52	50	47	43	38	36	35	34	32	30
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2032 2/	50	38	30	23	17	12	11	11	12	12	15
B. Bound Tests											
B1. Real GDP growth	50	52	50	47	43	38	36	35	34	32	30
B2. Primary balance	50	53	53	49	45	41	39	39	38	36	35
B3. Exports	50	76	120	112	104	94	90	88	85	79	76
B4. Other flows 3/	50	60	64	60	55	49	47	46	44	41	40
B5. Depreciation	50	52	33	30	26	22	21	21	21	19	18
B6. Combination of B1-B5	50	71	59	67	62	54	52	51	49	46	44
C. Tailored Tests											
C1. Combined contingent liabilities	50	59	59	56	54	50	49	50	50	49	49
Threshold	240	240	240	240	240	240	240	240	240	240	240
Debt service-to-exports ratio											
Baseline	3	4	4	4	4	4	6	6	6	6	5
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2032 2/	3	3	4	3	3	3	4	4	4	4	3
B. Bound Tests											
B1. Real GDP growth	3	4	4	4	4	4	6	6	6	6	5
B2. Primary balance	3	4	4	4	4	4	6	6	7	6	5
B3. Exports	3	4	7	7	6	8	12	13	13	12	11
B4. Other flows 3/	3	4	5	4	4	5	7	7	8	7	6
B5. Depreciation	3	4	4	3	3	4	4	5	5	5	4
B6. Combination of B1-B5	3	4	6	5	5	6	8	8	9	8	7
C. Tailored Tests											
C1. Combined contingent liabilities	3	4	5	4	4	4	6	7	7	6	6
Threshold	21	21	21	21	21	21	21	21	21	21	21
Debt service-to-revenue ratio											
Baseline	4	4	5	4	4	5	6	6	6	6	4
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2032 2/	4	4	4	4	3	3	4	4	4	3	2
B. Bound Tests											
B1. Real GDP growth	4	4	6	5	5	6	8	8	8	7	5
B2. Primary balance	4	4	5	5	4	5	7	7	7	6	5
B3. Exports	4	4	6	6	5	7	10	10	9	8	7
B4. Other flows 3/	4	4	5	5	4	6	8	8	7	7	5
B5. Depreciation	4	5	6	5	4	6	6	6	6	5	4
B6. Combination of B1-B5	4	4	6	5	5	7	8	8	8	7	6
C. Tailored Tests											
C1. Combined contingent liabilities	4	4	5	5	4	5	7	7	7	6	5
Threshold	23	23	23	23	23	23	23	23	23	23	23

Sources: Country authorities; and staff estimates and projections.
1/ A bold value indicates a breach of the threshold.
2/ Variables include real GDP growth, GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
3/ Includes official and private transfers and FDI.

Table 4. Moldova: Sensitivity Analysis for Key Indicators of Public Debt 2022–2032
(in percent)

	Projections 1/										
	2022	2023	2024	2025	2026	2027	2028	2029	2030	2031	2032
PV of Debt-to-GDP Ratio											
Baseline	32	33	34	35	34	34	33	32	30	29	28
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2032 2/	32	30	29	29	28	27	27	26	25	25	24
B. Bound Tests											
B1. Real GDP growth	32	39	50	56	62	66	70	**73**	**76**	**79**	**82**
B2. Primary balance	32	34	36	37	37	36	35	34	32	31	30
B3. Exports	32	37	46	47	46	45	43	41	39	37	35
B4. Other flows 3/	32	35	39	39	39	38	37	35	34	32	31
B5. Depreciation	32	32	32	32	31	29	28	26	24	22	20
B6. Combination of B1-B5	32	34	36	38	40	40	40	40	40	39	39
C. Tailored Tests											
C1. Combined contingent liabilities	32	44	45	45	44	43	42	40	38	37	35
TOTAL public debt benchmark	70	70	70	70	70	70	70	70	70	70	70
PV of Debt-to-Revenue Ratio											
Baseline	103	103	105	106	104	102	100	96	91	87	84
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2032 2/	103	95	92	89	86	83	82	79	75	73	71
B. Bound Tests											
B1. Real GDP growth	103	122	155	172	186	199	211	219	226	235	244
B2. Primary balance	103	107	114	113	111	109	106	102	96	93	89
B3. Exports	103	116	145	143	140	136	131	123	116	110	105
B4. Other flows 3/	103	111	121	121	119	116	112	107	101	96	92
B5. Depreciation	103	102	100	98	94	89	84	78	71	65	60
B6. Combination of B1-B5	103	106	113	118	120	121	122	120	118	117	117
C. Tailored Tests											
C1. Combined contingent liabilities	103	139	139	137	133	129	125	120	114	110	106
Debt Service-to-Revenue Ratio											
Baseline	5	10	13	15	19	23	27	28	27	27	27
A. Alternative Scenarios											
A1. Key variables at their historical averages in 2022-2032 2/	5	10	10	10	12	16	20	25	27	27	27
B. Bound Tests											
B1. Real GDP growth	5	11	19	29	41	52	62	67	71	73	77
B2. Primary balance	5	10	14	18	22	26	29	30	29	28	28
B3. Exports	5	10	13	16	20	25	30	31	30	29	29
B4. Other flows 3/	5	10	13	15	20	24	28	29	29	28	27
B5. Depreciation	5	10	13	13	18	22	26	27	26	25	25
B6. Combination of B1-B5	5	10	14	17	23	29	34	36	36	36	37
C. Tailored Tests											
C1. Combined contingent liabilities	5	10	27	27	32	33	35	35	34	32	31

Sources: Country authorities; and staff estimates and projections.

1/ A bold value indicates a breach of the benchmark.

2/ Variables include real GDP growth, GDP deflator and primary deficit in percent of GDP.

3/ Includes official and private transfers and FDI.